

The End of Programming as We Know It

Envisioning Radical Re-Conceptualizations of Co-Coding with AI

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Abstract

In this workshop we investigate "alternative futures" of programming with generative-AI tools. The workshop aims to establish a strand of HCI research on the future of AI-supported programming that goes beyond the current focus on usability and performance of GenAI-assisted programming tools, to instead explore radical re-conceptualizations of co-coding and co-creation with AI. The workshop is open to researchers and practitioners engaged in critically exploring novel ways of coding and interacting with AI-based programming tools. We encourage speculative design inquiries exploring prototypes, and provocations that facilitate engagement with a wider spectrum of human skills, experiences and ways of knowing when programming with AI. The workshop welcomes contributions that i) present examples of novel interaction, empirical studies or prototypes of AI-based programming, ii) re-conceptualize how programming with AI-based tools may be conducted and designed for, iii) reflect on values and ethical challenges of AI-based programming.

CCS Concepts

• **Human-centered computing** → **Interaction paradigms**; • **Software and its engineering** → **Software development techniques**;

Keywords

AI-based programming, co-creativity, co-coding, empirical study, Generative AI

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1 Motivation

The "new AI" in the form of Generative AI is having a profound impact on a broad range of activities that involve the production of original digital and interactive technology. We see how programming with GenAI-based is incorporated into professional tools and practices in software development and interaction design for a range of different purposes [4, 29]. The outstanding performance of several of these tools has even led to the claim that we are seeing "the end of programming" [31] in its current form, and that computational systems will predominantly be generated by AI systems rather than programmed by humans. This raises fundamental questions regarding whether this evolution actually broadens access to computational creativity or inadvertently narrows the spectrum

for the kinds of human skills and embodied knowledge that can be used in designing and building interactive systems.

These rapidly evolving capabilities of AI to produce high-quality programming code from natural language instructions have raised propositions that the future of programming will shift from being about writing syntactically and logically correct code to a process of articulating ideas and intentions from which the system could generate the code that one wishes. While such a view is philosophically problematic [26], studies of this form of AI-supported programming and design have also shown that the prompt and text-based approaches of articulating instructions are challenging for many users and that they limit the use of other kinds of skills and ways of knowing [1]. Here, HCI has a long tradition of investigating alternative understandings of programming [11], as well as exploring different modalities and interaction forms. However, the influx of Generative AI in practices of programming, prototyping and interaction design is still in its infancy, and HCI research needs to take on a critical role in further understanding and shaping the directions of this development in research, practice and education [14].

To engage with these challenges, the proposed workshop will open conversations around how HCI can envision alternative futures of interaction for programming with generative AI. The workshop is open to researchers and practitioners engaged in critically exploring and understanding novel ways of coding and interacting with AI-based programming tools in a fashion that go beyond traditional forms of articulating and expressing ideas and intentions in practices of programming. In particular, the workshop welcomes contributions that attempt to re-conceptualize how programming with AI-based tools may be conducted and the kinds of actions and activities they may be designed for. The workshop includes contributions that attempt to critically rethink the kind of interaction metaphors and frameworks around which tools and technologies for programming and coding are designed, and how they may be tailored to various users, settings and interests. At the core of these interests are also concerns of ethics, accountability and agencies that arise as human-AI relations in programming transform and evolve.

The workshop will entail activities inspired by approaches and methods of design fiction and speculative design to enable explorations beyond the predefined pathways, allowing for considerations novel forms of interaction and programming. At the core of such approaches lies their capacity to stimulate thinking beyond conventional development paradigms to allow for critical explorations and envisioning of future interactions, technologies, materials and settings. To investigate such a space of technologies we welcome contributions that critically reflect on the design and utilization of these technologies, as well as reflections on the consequences and ethics of their use.

1.1 Background

An increasing amount of studies has explored the effectiveness of novel AI tools such as Copilot for generating working programming code and how they integrate with and influence various aspects of the practices of programming, such as learning [4, 8], productivity[3], use [12], correctness and performance [5], etc. Bird, et al [3] for instance noted how programmers using Copilot spent less time on analysing code errors, but at the same time seemed to develop less of an understanding of how or why a particular piece of code worked the way it did. They also noted that the perceived productivity among users of Copilot was actually higher than the measured productivity. Other studies, (e.g. [19] [2]) show that users of Generative AI tools for programming were required to put additional efforts into actions such as instructing and articulating how the code should work, as well as on reading, reviewing, debugging and tweaking the code that had been generated, while less efforts were put into spending time on writing actual lines of code.

This development has been argued as pointing to a potential larger shift in the nature of future practices of programming, making programming less about writing code from scratch, and more about being able to interpret, understand and tweak code generated by AI systems. This has been argued to suggest that the future of programming will change from programming being a process of writing syntactically and logically correct lines of code, to a process of being skilled in expressing and instructing the system in ways that makes it generate the code that one wants. A few recent examples of studies have explored how to design for novel kinds of programming with Generative AI may look beyond current interaction models, for instance, through prototypes that use the ability of ChatGPT to generate code based on visual and textual sketches of a program, e.g., [1, 17] and how to create programming assistance through conversational interactions when programming using LLMs [21]. Few of these growing number of studies on the usage of generative AI for programming have stretched beyond traditional conceptions of programming to explore specific properties of the user interfaces, and the opportunities for alternative forms of programming. In the proposed workshop, we aim to open up for and start exploring design of AI-based co-programming in ways that go beyond the now predominant text and chat-based forms of interaction with these tools.

These types of ongoing shifts in programming practices have been argued to fundamentally recast the relations between programmers and the technologies they are entangled with in the creation of programming code. Both Welsch [31] and Jonsson & Tholander [9] point to how these novel forms of programming require users to give up autonomy and hand over control over parts of the coding and creative process to the AI tool [3]. This points to a need to design ways of enabling new forms of partnerships and ways of interacting between human and artificial actors, in which the AI is considered more of an autonomous collaborator. This aligns with notions of co-creativity and co-creation that have emerged in HCI [30], which frame how interactive and smart technologies may be considered as co-participants in creative processes, rather than as mere tools controlled by human actors. This further relates to research around notions such as more-than-human design, co-performance and machine agency [15] that reconsiders the view of humans as the sole source of creative agency. Instead, these theories highlight the entanglement of human agency with agencies originating from non-human entities, implying that programming and design are inherently *relational* forms of action. Programming and design are then not to be viewed as exclusively human actions but co-constituted in assemblies of human and *artificial agencies*. These theoretical underpinnings challenge how alternative metaphors,

relations and forms of interaction in AI-based programming are envisioned, as well as open up for critical exploration of alternative values, ethics and ways of knowing in the emergence of creative practices involving both humans and AI technology. These perspectives highlight a number of concerns that will be addressed in the workshop:

The integration of natural language and code - LLMs are largely based on language communication. This raises questions on how to refine natural language interaction, potentially requiring a *reconceptualisation of software* where code and natural language expressions are considered integral parts of software. If software is produced through natural language interfaces in real-time, are special-purpose applications needed at all?

Collaboration and co-coding - Most AI-assisted programming tools are designed around individual use. How can collaborative and socially situated forms of programming be designed for? Examples include how new tools may allow for mixed media for collaborative expressions, e.g. to incorporate live video recordings and hand-drawn sketches as shareable resources in the programming practice [22].

Complexity and control – The expansive development of AI-based programming tools will potentially automate an increasing part of the creation process, leading to the generation of large amounts of code that may become challenging for programmers to oversee and comprehend. Live coding with AI tools provides one such example where the code writing and code generation process is an integral part of the final executable outcome. This raises issues of the design of novel interfaces, representations, and visualisations to make code interpretable and explainable, as well as how the programmer can maintain control of what is generated and what it expresses. For instance, how may AI tools support the reading and understanding of code by translating it to spoken explanations in natural language, animations or other forms of representations?

Craftsmanship, creativity, and skilled practice - AI tools that automate parts of the programming process might encourage over-reliance on the capabilities of the tools. This stands in contrast with perspectives of programming that draw on craftsmanship, attending to details and uniqueness in the creative process, and keeping ownership of the development process [10].

1.2 Workshop themes and topics of interest

Authors may wish to address, but are not limited to, the following three broad themes:

(1) **Novel interactions and practices of AI-based programming**
   - Empirical studies of new forms and practices of programming with Generative-AI tools?
   - Prototypes for programming with Generative-AI beyond text-based forms of representation and expression?
   - Methods and approaches to envision tools and interactions that go beyond current AI-based programming.
   - Guiding principles for moving towards programming practices where alternative skills, forms of expression, and ways of knowing are used?

(2) **Understanding and conceptualising relations in human-AI coding and programming**
   - Managing and orchestrating agency and accountability, and more-than-human perspectives in human AI-collaborative programming practices
   - Balancing expressivity, creativity, control, and automation in Generative-AI programming tools
   - Emerging types of coding practices and user needs
   - Craft perspectives on Generative-AI co-creation

(3) **Critical perspectives on AI based programming**
   - Ethical concerns of AI-based code-generation practices
   - Human values and bias in automated code generation
   - Responsible development of AI-based programming tools

1.3 Workshop goals

An overarching goal with the workshop is the formation of a community of researchers with a particular interest in aspects related to Generative-AI based programming. This specifically entails: i) bringing together researchers with different perspectives and initiate conversations around novel directions of programming interfaces and practices, ii) fostering conversations around challenges related to generative AI in programming and provide a starting point for critical explorations of programming and HCI, iii) collecting examples of new concepts, practices and metaphors for analyzing and understanding generative AI in programming, and iv) creating bridges to other relevant HCI research research areas on human-AI interaction, visual programming, etc.

2 Organizers

The organization committee consists of researchers from eight different universities and organisations, and who provide different perspectives and relevant experiences to the suggested topic.

Martin Jonsson is a professor in Media Technology at Södertörn University. Martin has been engaged in several research projects exploring novel interaction forms and embodied interaction as well as research on making and programming practices, and is currently PI for a project on Co-coding with AI.

Jakob Tholander is a professor in Human-Computer Interaction at the Department of Computer and Systems Science at Stockholm University. Jakob has been involved in a number of research projects, including tangible and embodied interaction and sustainable interaction, and is currently working in a project on Co-coding with AI.

Mattias Rost is a senior lecturer in Interaction Design at Gothenburg University. Mattias is investigating novel uses of GenAI in interaction design, in particular non-text based UIs. He also has an interest in programming education using AI assistants.

Ylva Fernaeus is associate professor in Interaction Design at KTH as well as at Umeå Institute of Design, and coordinator of the AI and Art focus area at TAIGA. Ylva has worked with programming in the context of children's games, tangible programming systems, robotic devices, and embodied understandings of everyday programming practice.

Advait Sarkar is a researcher at Microsoft, affiliated lecturer at the University of Cambridge, and honorary lecturer at University College London. He studies the effects of Generative AI on knowledge work [23, 28], programming [16, 24], and data analysis [6, 7, 13, 18]. He is part of the *Tools for Thought* group at Microsoft

[20] and leads a research agenda aimed on critical thinking with Generative AI [26, 27]. His article *"AI Should Challenge, Not Obey"* appears in the October 2024 issue of Communications of the ACM [25].

Ahmed Kharrufa is a reader in AI for Education at the School of Computing, Newcastle University. Ahmed is currently involved in several projects exploring the potential and implications of Generative AI on education in general, and computing science education in particular. He is particularly interested in the use of Generative AI in collaborative work, programming and HCI work practices.

Thomas Ludwig is a professor in Educational Technologies for Digital Transformation at the FernUniversität in Hagen. Thomas heads the interdisciplinary research focus "Work - Education - Digitalization" and is responsible for the scientific-technical working group of the Future Center of Artificial Intelligence. His main research interests are the human-centered design of assistance systems, artificial intelligence, and immersive learning.

Tobias Ahlin Bjerrome is a Principal Design Engineer at GitHub, and part of the GitHub Copilot team, with previous work experience as Experience Design Director at Mojang and UI Design Lead at Spotify. Tobias gives public lectures on AI, creativity and design.

3 Plans to Publish Workshop Proceedings

We aim to publish the workshop position papers as a collection in ArXiv given the consent of the authors

4 In-Person or Hybrid

The workshop will be conducted in a hybrid format, encouraging *on-site* presence, but open up for online participation in cases where on-site participation is not possible.

5 Workshop Design

5.1 Pre-workshop plan

The call for participation and other workshop-related information will be made available on a workshop website. We will circulate the workshop call through the organizers' professional networks, SIGCHI and other relevant mailing lists, and social media. Authors are invited to submit a 2-4 page position paper following the ACM single-column review format as a pdf-file. We aim to promote broader participation, thus offering various ways displaying research findings and ideas that may include design examples, conceptual reflections, provocations, and design fictions. We encourage authors to explicitly address one of the workshop themes. Contributions may combinations of design cases, prototype descriptions, empirical results, or proposals for future research. The submissions will be reviewed by the workshop organizers and selected based on their quality, novelty, relevance to the workshop's topics with a particular focus on the possibility to open up for reflection and conversation. We will aim for a balance in different contributions. Authors' will be asked for consent to publish that papers on the workshop website prior to the workshop to allow participants to read each contribution. At least one author of each accepted paper must attend the workshop' and register for it. We will do our best to accommodate specific accessibility needs with the help from the conference organizers. Authors are invited to specify any particular requirement in their position papers, or by directly contacting the organizers.

TIME	ACTIVITY
9.00 – 9.20	Welcome and introduction to the workshop
9.20 – 10.40	Presentation of position papers – possibly in an alternative format to open up for questions and discussion
10.40 – 11.00	Coffe break
11.00 - 12.15	Group-based design activity to envision, sketch, reflect and evaluate various themes of the future of programming. Activities will open up for making use of the outdoor areas around the convention centre, bringing in different materials and encourage participants to bring in their various backgrounds and experiences into the conversations.
12.15 - 13.30	Lunch
13.30 - 14.00	Design activity continued
14.00 – 14.30	Shared critiquing and reflection on outcome thus far
14.30 – 15.00	Second design activity based on input and insights from reflections
15.00 – 15.30	Coffee
15.30 – 16.15	Synthetising and articulating outcomes of design activity
16.15 – 17.00	Presentations
17.00 – 17.30	Final reflections and future plans

Table 1: Preliminary Workshop Schedule

5.2 Key dates

The workshop's planned key dates are:

- June 16: submission deadline for authors.
- June 26: notification of acceptance
- August 19: Workshop at Critical Computing 2025

5.3 Workshop activities

The workshop is organized as a one-day event with a focus on fostering conversations among the participants. A preliminary workshop schedule is outlined in Table 1. The workshop will include an initial round of rapid presentations of the position papers, followed by the main design-oriented activity revolving around the challenges connected to the research on the future of programming with AI. The workshop will be designed to avoid too much time being given to individual presentations, and instead emphasize collaboration and creative and critical idea generation. Based on the participants' position papers we will formulate a number of specific themes that will be used as starting points for the design activities. We believe that this will provide opportunities for everyone to engage in a theme that is close to their specific research interest. The workshop will be set up as an on-site activity; however, remote participation will be possible but with limited (or individual) participation in the design activities.

5.4 Post-workshop plans

The workshop aims at building community between researchers and practitioners with an interest in generative AI in programming. The workshop will act as an initial step for fostering long-term collaborations, including the organization of future events (e.g., organizing research workshops or other joint research activities). We aim to write a piece for ACM Interactions to outline the most challenging and inspirational outcomes of the workshop and point to future opportunities for joint research. Participants' suggestions on possible future actions will also be discussed at the workshop. Given the participants' consent, the position papers will remain available at the workshop's website.

6 Call for Participation

In "the end of programming"-workshop we will investigate the potential "alternative futures" of programming with generative AI tools. The workshop aims to establish a strand of HCI research on the future of AI-supported programming that goes beyond the current focus on increasing the usability and performance of GenAI-assisted programming tools, to instead explore radical reconceptualizations of co-coding and co-creation with AI. The workshop is open to researchers and practitioners engaged in critically exploring and understanding novel ways of coding and interacting with AI-based programming tools in a fashion that goes beyond traditional forms of articulating and expressing ideas and intentions in practices of programming.

We encourage speculative design inquiries exploring prototypes, interaction forms and provocations that facilitate engagement with a wider spectrum of human skills, experiences and ways of knowing when programming and coding with AI. The workshop welcomes contributions that i) present examples of novel interaction, empirical studies or prototypes of AI-based programming, ii) reconceptualize how programming with AI-based tools may be conducted and the actions and activities they are designed for, iii) reflect on values and ethical challenges of AI-based programming.

Authors are invited to submit a 2-4 page position paper following the ACM single-column review format as a pdf-file. We promote broader participation including various ways of displaying research findings and ideas, including design examples, conceptual reflections, provocations, and design fictions. We encourage authors to explicitly address one of the workshop themes. The submissions will be reviewed by the workshop organizers and selected based on their relevance to the workshop's themes and the possibility to open up for reflection and conversation.

Position papers to be sent to: jakobth@dsv.su.se.

- Deadline: **June 16, 2025**.
- Notification of Acceptance: **June 26, 2025**.
- More information: https://mi.sh.se/ shmnjn07/theend

7 Expected size of attendance

We plan for a maximum of 30 in-person attendees, and will open up for online participants for authors who cannot attend physically. For submissions with several authors, will we ask how many persons wish to attend.

References

[1] Seonghoon Ban and Kyung Hoon Hyun. 2020. 3D Computational Sketch Synthesis Framework: Assisting Design Exploration Through Generating Variations of User Input Sketch and Interactive 3D Model Reconstruction. *Computer-Aided Design* 120, C (March 2020). https://doi.org/10.1016/j.cad.2019.102789

[2] Shraddha Barke, Michael B. James, and Nadia Polikarpova. 2022. Grounded Copilot: How Programmers Interact with Code-Generating Models. http://arxiv.org/abs/2206.15000 arXiv:2206.15000 [cs].

[3] Christian Bird, Denae Ford, Thomas Zimmermann, Nicole Forsgren, Eirini Kalliamvakou, Travis Lowdermilk, and Idan Gazit. 2023. Taking Flight with Copilot. *Commun. ACM* 66, 6 (2023), 56–62. https://doi.org/10.1145/3589996

[4] C Bull and A Kharrufa. 2023. Generative AI Assistants in Software Development Education: A vision for integrating Generative AI into educational practice, not instinctively defending against it. *IEEE Software* (2023).

[5] Paul Denny, Viraj Kumar, and Nasser Giacaman. 2023. Conversing with Copilot: Exploring Prompt Engineering for Solving CS1 Problems Using Natural Language. In *Proceedings of the 54th ACM Technical Symposium on Computer Science Education V. 1 (SIGCSE 2023)*. Association for Computing Machinery, New York, NY, USA, 1136–1142. https://doi.org/10.1145/3545945.3569823

[6] Ian Drosos, Advait Sarkar, Xiaotong Xu, Carina Negreanu, Sean Rintel, and Lev Tankelevitch. 2024. "It's like a rubber duck that talks back": Understanding Generative AI-Assisted Data Analysis Workflows through a Participatory Prompting Study. In *Proceedings of the 3rd Annual Meeting of the Symposium on Human-Computer Interaction for Work* (Newcastle upon Tyne, United Kingdom) *(CHIWORK '24)*. Association for Computing Machinery, New York, NY, USA, Article 16, 21 pages. https://doi.org/10.1145/3663384.3663389

[7] Kasra Ferdowsi, Jack Williams, Ian Drosos, Andrew D. Gordon, Carina Negreanu, Nadia Polikarpova, Advait Sarkar, and Benjamin Zorn. 2023. COLDECO: An End User Spreadsheet Inspection Tool for AI-Generated Code. In *2023 IEEE Symposium on Visual Languages and Human-Centric Computing (VL/HCC)*. 82–91. https://doi.org/10.1109/VL-HCC57772.2023.00017

[8] Johan Jeuring, Roel Groot, and Hieke Keuning. 2023. What Skills Do You Need When Developing Software Using ChatGPT? (Discussion Paper). In *Proceedings of the 23rd Koli Calling International Conference on Computing Education Research*. ACM, Koli Finland, 1–6. https://doi.org/10.1145/3631802.3631807

[9] Martin Jonsson and Jakob Tholander. 2022. Cracking the code: Co-coding with AI in creative programming education. In *Proceedings of the 14th Conference on Creativity and Cognition (C&C '22)*. Association for Computing Machinery, New York, NY, USA, 5–14. https://doi.org/10.1145/3527927.3532801

[10] Martin Jonsson and Jakob Tholander. 2024. Craft Ethics - Aiming for Virtue in Programming with Generative AI. In *Proceedings of the 35th Annual Conference of the Psychology of Programming Interest Group (PPIG 2024)*.

[11] Martin Jonsson, Jakob Tholander, and Ylva Fernaeus. 2009. Setting the stage – Embodied and spatial dimensions in emerging programming practices. *Interacting with Computers* 21, 1-2 (Jan. 2009), 117–124. https://doi.org/10.1016/j.intcom.2008.10.004 Conference Name: Interacting with Computers.

[12] Majeed Kazemitabaar, Xinying Hou, Austin Henley, Barbara Jane Ericson, David Weintrop, and Tovi Grossman. 2023. How Novices Use LLM-based Code Generators to Solve CS1 Coding Tasks in a Self-Paced Learning Environment. In *Proceedings of the 23rd Koli Calling International Conference on Computing Education Research*. ACM, Koli Finland, 1–12. https://doi.org/10.1145/3631802.3631806

[13] Majeed Kazemitabaar, Jack Williams, Ian Drosos, Tovi Grossman, Austin Henley, Carina Negreanu, and Advait Sarkar. 2024. Improving Steering and Verification in AI-Assisted Data Analysis with Interactive Task Decomposition. *arXiv preprint arXiv:2407.02651* (2024).

[14] Ahmed Kharrufa and Ian Johnson. 2024. The Potential and Implications of Generative AI on HCI Education. In *Proceedings of the 6th Annual Symposium on HCI Education*. 1–8.

[15] Lenneke Kuijer and Elisa Giaccardi. 2018. Co-performance: Conceptualizing the Role of Artificial Agency in the Design of Everyday Life. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, USA, 1–13. https://doi.org/10.1145/3173574.3173699

[16] Michael Jing Long Lee, Advait Sarkar, and Alan F. Blackwell. 2024. Predictability of identifier naming with Copilot: A case study for mixed-initiative programming tools. In *Proceedings of the 35th Annual Conference of the Psychology of Programming Interest Group (PPIG 2024)*.

[17] Clayton Lewis. 2019. Why can't programming be like sketching?. In *Proceedings of the Conference Companion of the 3rd International Conference on Art, Science, and Engineering of Programming*. ACM, Genova Italy, 1–6. https://doi.org/10.1145/3328433.3338060

[18] Michael Xieyang Liu, Advait Sarkar, Carina Negreanu, Benjamin Zorn, Jack Williams, Neil Toronto, and Andrew D. Gordon. 2023. "What It Wants Me To Say": Bridging the Abstraction Gap Between End-User Programmers and Code-Generating Large Language Models. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) *(CHI '23)*. Association for Computing Machinery, New York, NY, USA, Article 598, 31 pages. https:

//doi.org/10.1145/3544548.3580817

[19] James Prather, Brent N. Reeves, Paul Denny, Brett A. Becker, Juho Leinonen, Andrew Luxton-Reilly, Garrett Powell, James Finnie-Ansley, and Eddie Antonio Santos. 2023. "It's Weird That it Knows What I Want": Usability and Interactions with Copilot for Novice Programmers. *ACM Transactions on Computer-Human Interaction* 31, 1 (Nov. 2023), 4:1–4:31. https://doi.org/10.1145/3617367

[20] Microsoft Research. 2024. Tools for Thought. https://www.microsoft.com/en-us/research/project/t4t/. Accessed: 2024-09-26.

[21] Steven I. Ross, Fernando Martinez, Stephanie Houde, Michael Muller, and Justin D. Weisz. 2023. The Programmer's Assistant: Conversational Interaction with a Large Language Model for Software Development. In *Proceedings of the 28th International Conference on Intelligent User Interfaces*. ACM, Sydney NSW Australia, 491–514. https://doi.org/10.1145/3581641.3584037

[22] Geert Jacob Roumen and Ylva Fernaeus. 2021. Envisioning Arduino Action: A collaborative tool for physical computing in educational settings. *International Journal of Child-Computer Interaction* 29 (Sept. 2021), 100277. https://doi.org/10.1016/j.ijcci.2021.100277

[23] Advait Sarkar. 2023. Exploring Perspectives on the Impact of Artificial Intelligence on the Creativity of Knowledge Work: Beyond Mechanised Plagiarism and Stochastic Parrots. In *Proceedings of the 2nd Annual Meeting of the Symposium on Human-Computer Interaction for Work* (Oldenburg, Germany) *(CHIWORK '23)*. Association for Computing Machinery, New York, NY, USA, Article 13, 17 pages. https://doi.org/10.1145/3596671.3597650

[24] Advait Sarkar. 2023. Will Code Remain a Relevant User Interface for End-User Programming with Generative AI Models?. In *Proceedings of the 2023 ACM SIGPLAN International Symposium on New Ideas, New Paradigms, and Reflections on Programming and Software* (Cascais, Portugal) *(Onward! 2023)*. Association for Computing Machinery, New York, NY, USA, 153–167. https://doi.org/10.1145/3622758.3622882

[25] Advait Sarkar. 2024. AI Should Challenge, Not Obey. *Commun. ACM* (Sept. 2024), 5 pages. https://doi.org/10.1145/3649404 Online First.

[26] Advait Sarkar. 2024. Intention Is All You Need. In *Proceedings of the 35th Annual Conference of the Psychology of Programming Interest Group (PPIG 2024)*.

[27] Advait Sarkar, Xiaotong (Tone) Xu, Neil Toronto, Ian Drosos, and Christian Poelitz. 2024. When Copilot Becomes Autopilot: Generative AI's Critical Risk to Knowledge Work and a Critical Solution. In *Proceedings of the Annual Conference of the European Spreadsheet Risks Interest Group (EuSpRIG 2024)*.

[28] Lev Tankelevitch, Viktor Kewenig, Auste Simkute, Ava Elizabeth Scott, Advait Sarkar, Abigail Sellen, and Sean Rintel. 2024. The Metacognitive Demands and Opportunities of Generative AI. In *Proceedings of the CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) *(CHI '24)*. Association for Computing Machinery, New York, NY, USA, Article 680, 24 pages. https://doi.org/10.1145/3613904.3642902

[29] Jakob Tholander and Martin Jonsson. 2023. Design Ideation with AI - Sketching, Thinking and Talking with Generative Machine Learning Models. In *Proceedings of the 2023 ACM Designing Interactive Systems Conference (DIS '23)*. Association for Computing Machinery, New York, NY, USA, 1930–1940. https://doi.org/10.1145/3563657.3596014

[30] Ron Wakkary. 2021. *Things We Could Design: For More Than Human-Centered Worlds*. MIT Press.

[31] Matt Welsh. 2023. The End of Programming. *Commun. ACM* 66, 1 (Jan. 2023), 34–35. https://doi.org/10.1145/3570220